ZS

Virality Prediction in Online Social Networks:- A Novel Approach

Akrati Saxena, Yayati Gupta

November 12, 2014

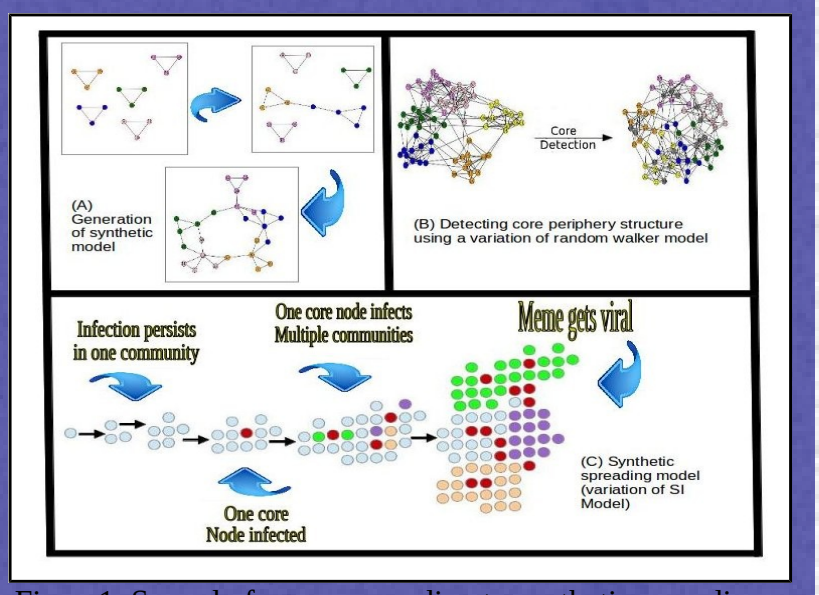

Figure1: Spread of meme according to synthetic spreading model

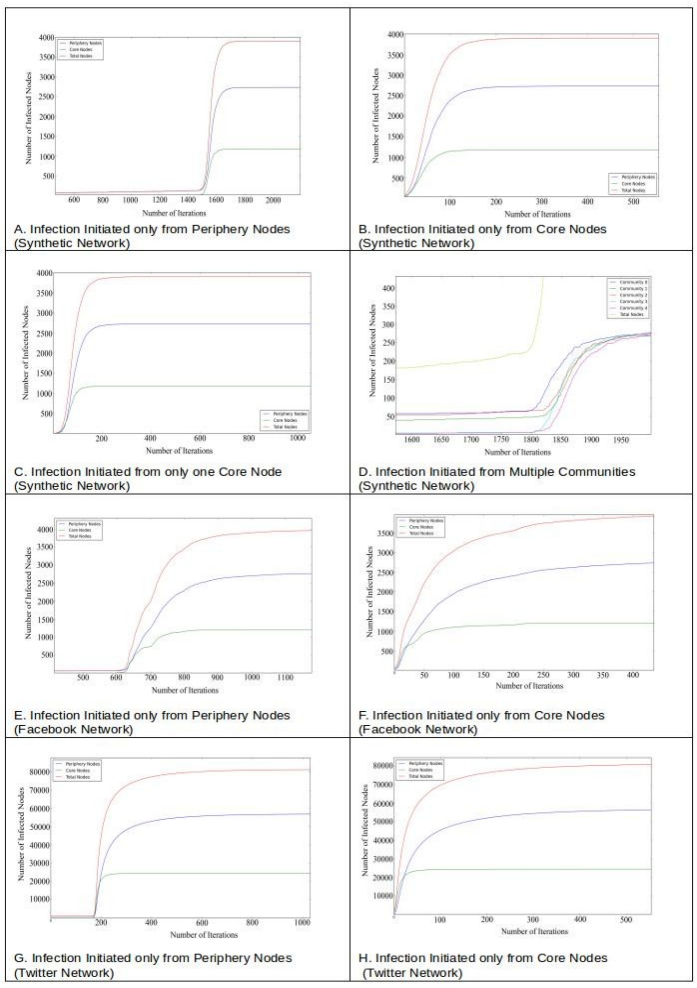

Figure 2: Spreading pattern according to different initial distributions of infection

Introduction

A meme is information that is seen and shared by the users of a social networking site. If a large fraction of online population adopts a meme, sees and shares it, the meme is called viral. There are many applications of predicting the virality of a meme. The popularity of a person can be predicted by looking at the number of views of an information shared by him on the internet. Many other applications of this study are predicting the spread of epidemic in a population[1], predicting stock market indicators[2], viral marketing, crime prediction etc. Many approaches have been taken till date for the prediction of virality. The most elementary studies in this direction consisted of analysing the content of a meme like its nature, importance and the intensity of emotions[3] it invokes in its audience. Other studies focus on the structure of the network and the kind of people who are spreading the information[4][5]. In our study, we analyse the virality of a meme in a community structure using the concept of core.

Important Definitions

A network is said to be **scale free** if its degree distribution follows a power law ie. fraction of nodes having degree k, $P(k) \sim k^{(-\gamma)}$, where $2 < \gamma < 3$. Hence, such networks can be characterized by the presence of hubs, but only a few. A network is said to possess **community structure** if it can be divided into sets of nodes called communities, the nodes in which are densely connected internally but not externally. A network is said to possess **core periphery structure** if its nodes can be divided in two sets - core and periphery. The nodes in the set core are densely connected to whole network. The nodes in the periphery set are connected densely externally but not internally.

Our Approach

The synthetic model employed in our study has been generated using the algorithm described in [6] . The networks generated in this manner have a number of communities and a scale free structure. Then, we detect the core in our network using a variation of the random walker model described in [7]. Core nodes constitute a community of their own. So, a network with a number of communities is obtained with a core periphery structure intertwined in it.

A variation of SI spreading model is used for spreading a meme in a network. The probability of a meme getting spread from a node A to node B (connected through an edge) varies in accordance with the types of nodes A and B. Core nodes are able to infect their neighbours very easily. Periphery nodes can infect the peripheral nodes of same community easily but not of different community. In addition, it is very tough for a periphery node to infect a core node.

The impact of synthetic spreading model is seen on the synthetic as well as real world network. Different cases for virality are obtained starting from the different distributions of the initially infected nodes.

Results

The spreading pattern has been observed for synthetic networks having approximately 30,000 to 60,000 nodes as well as for the real world social networks- facebook and twitter. It is observed that the virality of a meme accelerates sharply as soon as a sufficient proportion of core nodes are infected. Also, Once a core node gets infected, it does not take much time for a sufficient fraction of them to be infected. Also, the probability of a meme spreading in multiple communities increases as it hits atleast one core node.

Conclusion

We show that the virality of a meme increases drastically on hitting any of the core nodes in the network. We performed our study using synthetic networks of approximately one million nodes and a variant of SI disease spreading model. We derived the analytic expressions for the probability of a meme going viral by looking at its initial spread. It is shown that the core nodes play a very important role in the spread of information on any social networking site. The results are also verified on twitter and facebook data. Future work in this direction may include the use of more complex spreading models. Also, a more accurate algorithm for core detection can be used. Our method can be extended by doing a detailed analysis of the spread of meme in communities.

References:

[1] Aron Culotta , Towards detecting influenza epidemics by analyzing Twitter messages, Proceedings of the First Workshop on Social Media Analytics Pages 115-122, 2010

[2] Xue Zhang1,Hauke Fuehres,Peter A. Gloor, *Predicting Stock Market Indicators Through Twitter "I hope it is not as bad as I fear", Procedia - Social and Behavioral Sciences, 2011.*

[3] J. Berger, K. Milkman, S*ocial Transmission, Emotion, and the Virality of Online Content, Wharton Research Paper, 2010.*

[4]L Weng, F Menczer, YY Ahn , Virality prediction and community structure in social networks, Scientific Reports, 2013.

[5] M Kitsak, LK Gallos, S Havlin, F Liljeros, *Identification of influential spreaders in complex networks, Nature Physics, 2010.*

[6]*J. Wu, Z.Gao, H. Sun , Cascade and breakdown in scale-free networks with community structure, Physical Review E, 2006.spreaders in complex networks, Nature Physics, 2010.*

[7]*Fabio Della Rossa, Fabio Dercole, Carlo Piccardi.* Profiling core-periphery network structure by random walkers, Scientific Reports, 2013